

MAINTENANCE OPERATIONS CLIENT

User Stories & Acceptance Criteria

Maintenance Operations Analytics & Reporting Automation

Document Control	Detail
Engagement	Aug–Nov 2024
Role	Business Analyst + Data Analyst
Primary User / Stakeholder	Maintenance Manager
Data Scope	Approximately 10,000 historical records across 20 teams
Related Artifact	Business Requirements Document (BRD)
Status	Portfolio reconstruction based on confirmed engagement facts

Integrity note: These stories formalize confirmed business needs for portfolio presentation. They are not represented as verbatim historical Jira tickets or formally signed client acceptance records.

1. Purpose & Story Standard

The user stories translate the Maintenance Manager's reporting needs into testable analytical and dashboard requirements. Each story follows the structure: As a [user], I want [capability], so that [business value].

Primary persona: Maintenance Manager

Confirmed rules: $\text{Completion Rate} = \frac{\text{Completed Maintenance Tasks}}{\text{Total Assigned Maintenance Tasks}} \times 100$. Overdue = Due Date < Current Date AND Status ≠ Completed.

2. User Story Catalogue

ID	Story	Primary Requirement Trace
US-01	Consolidated Maintenance Overview	BR-01, BR-02, BR-16, BR-17
US-02	Work Order Status & Completion	BR-02, BR-03, BR-04, BR-05
US-03	Overdue Work Monitoring	BR-06
US-04	Completion-Time Analysis	BR-07
US-05	Priority & High-Priority Pending Work	BR-08, BR-13
US-06	Team Performance	BR-09, BR-12
US-07	Technician Performance	BR-10
US-08	Backlog & Workload Visibility	BR-11, BR-12
US-09	Maintenance Category & Asset Analysis	BR-14, BR-15
US-10	Interactive Power BI Reporting	BR-16, BR-17, BR-18
US-11	Reliable & Confidential Reporting	BR-01 through BR-18; NFR-01 through NFR-06

US-01 — Consolidated Maintenance Overview

User Story: As a Maintenance Manager, I want a consolidated view of maintenance operations so that I can understand overall work status without manually combining multiple reports.

Acceptance Criteria

- AC-01.1: The solution presents maintenance information from the prepared multi-source dataset in a consolidated Power BI reporting layer.
- AC-01.2: The overview provides visibility into assigned/completed work and relevant operational status measures.
- AC-01.3: Completion Rate follows the confirmed calculation rule.
- AC-01.4: The portfolio may show the confirmed 92% overall completion rate as an observed analytical finding.
- AC-01.5: The 92% figure must not be described as an improvement caused by the project.
- AC-01.6: Equivalent headline KPIs must reconcile to the prepared analytical data.

BRD Trace: BR-01, BR-02, BR-16, BR-17

US-02 — Work Order Status & Completion

User Story: As a Maintenance Manager, I want to monitor completed, open and pending maintenance work so that I can understand execution status and outstanding workload.

Acceptance Criteria

- AC-02.1: Maintenance work can be analyzed by standardized status categories.
- AC-02.2: Completed work is reported consistently with the completion-rate logic.
- AC-02.3: Open and pending work can be viewed without assuming they are interchangeable unless the reconstructed status mapping explicitly defines them that way.
- AC-02.4: Status standardization is documented and tested.
- AC-02.5: Unknown or ambiguous source statuses are not silently forced into a valid category.
- AC-02.6: Status totals reconcile to the eligible prepared records.

BRD Trace: BR-02, BR-03, BR-04, BR-05

US-03 — Overdue Work Monitoring

User Story: As a Maintenance Manager, I want to identify overdue maintenance work so that I can focus attention on tasks that have passed their due date and remain incomplete.

Acceptance Criteria

- AC-03.1: An item is classified as overdue when Due Date < Current Date AND Status ≠ Completed.
- AC-03.2: Records without a valid due date are not falsely classified as overdue.
- AC-03.3: Completed items are excluded from overdue work under the confirmed rule.
- AC-03.4: Overdue work can be analyzed alongside relevant dimensions such as team, technician, priority, category or asset where supported.
- AC-03.5: Overdue counts shown in the portfolio must come from sanitized/reconstructed data rather than invented historical client values.

BRD Trace: BR-06

US-04 — Completion-Time Analysis

User Story: As a Maintenance Manager, I want to analyze maintenance completion time so that I can understand how long completed work takes and investigate delays.

Acceptance Criteria

- AC-04.1: Average completion time is calculated only for records with valid dates required by the documented rebuilt rule.
- AC-04.2: The calculation convention used in the portfolio is documented.
- AC-04.3: Invalid or ambiguous date records do not silently distort the measure.
- AC-04.4: Completion-time analysis can be filtered by relevant operational dimensions where supported.
- AC-04.5: The portfolio does not claim an exact historical average completion-time value that is not retained.

BRD Trace: BR-07

US-05 — Priority & High-Priority Pending Work

User Story: As a Maintenance Manager, I want maintenance work broken down by priority and want visibility into high-priority pending items so that critical outstanding work can be identified.

Acceptance Criteria

- AC-05.1: Maintenance records can be grouped by standardized priority values where source data supports them.
- AC-05.2: High-priority pending work is visible as an exception area.
- AC-05.3: Missing/unknown priority values remain identifiable rather than being assigned an unsupported priority.
- AC-05.4: Priority filtering does not change the underlying definitions of completion or overdue work.
- AC-05.5: No unsupported historical priority counts or percentages are presented as client facts.

BRD Trace: BR-08, BR-13

US-06 — Team Performance

User Story: As a Maintenance Manager, I want to compare maintenance performance across teams so that I can identify differences in completion and workload.

Acceptance Criteria

- AC-06.1: The prepared data uses standardized team names.
- AC-06.2: Team-level views support relevant measures such as work volume and completion.
- AC-06.3: The analysis can expose differences in team performance.
- AC-06.4: Workload comparisons use documented measures rather than an invented historical imbalance threshold.
- AC-06.5: The portfolio does not fabricate exact historical team rankings or team-level percentages.

BRD Trace: BR-09, BR-12

US-07 — Technician Performance

User Story: As a Maintenance Manager, I want technician-level maintenance visibility so that I can understand activity and performance across technicians.

Acceptance Criteria

- AC-07.1: Technician-level reporting uses available technician attribution from prepared data.
- AC-07.2: Technician performance is evaluated using relevant operational measures rather than an unsupported single composite score.
- AC-07.3: Missing technician attribution is handled transparently.
- AC-07.4: Filters and aggregations preserve KPI consistency.
- AC-07.5: The portfolio does not fabricate exact historical technician rankings or performance scores.

BRD Trace: BR-10

US-08 — Backlog & Workload Visibility

User Story: As a Maintenance Manager, I want to monitor maintenance backlog and workload distribution so that I can identify accumulated work and operational imbalance.

Acceptance Criteria

- AC-08.1: Backlog is defined explicitly before it is calculated in the portfolio reconstruction.
- AC-08.2: The exact historical backlog definition is not claimed because it is not retained.
- AC-08.3: Backlog can be analyzed across relevant operational dimensions where supported.
- AC-08.4: Workload imbalance is assessed using a documented comparison method.
- AC-08.5: No unsupported backlog size, threshold or team imbalance percentage is presented as historical fact.

BRD Trace: BR-11, BR-12

US-09 — Maintenance Category & Asset Analysis

User Story: As a Maintenance Manager, I want to analyze recurring maintenance categories and asset/equipment patterns so that I can identify areas requiring operational attention.

Acceptance Criteria

- AC-09.1: Maintenance categories are standardized sufficiently for grouping and comparison.
- AC-09.2: Recurring categories can be identified using frequency/trend analysis.
- AC-09.3: Asset/equipment-level reporting uses available asset/equipment attribution.
- AC-09.4: The analysis may identify equipment-related patterns/issues but does not infer unsupported failure causation.
- AC-09.5: Exact historical category frequencies and asset rankings are not fabricated.

BRD Trace: BR-14, BR-15

US-10 — Interactive Power BI Reporting

User Story: As a Maintenance Manager, I want an interactive Power BI dashboard so that I can explore maintenance performance without rebuilding weekly reports manually.

Acceptance Criteria

- AC-10.1: The dashboard is built on prepared maintenance data.
- AC-10.2: Relevant filters update intended visuals consistently.
- AC-10.3: Equivalent KPI definitions remain consistent across dashboard views.
- AC-10.4: Dashboard views support the confirmed analytical areas.
- AC-10.5: The solution is understandable through supporting documentation/training.
- AC-10.6: The confirmed engagement benefit may be stated as approximately 3–4 hours of weekly manual reporting effort eliminated.

BRD Trace: BR-16, BR-17, BR-18

US-11 — Reliable & Confidential Reporting

User Story: As a Maintenance Manager, I want the dashboard to use reliable and controlled data so that decisions are not based on duplicate, inconsistent or misleading records.

Acceptance Criteria

- AC-11.1: Duplicates, missing values, inconsistent team names, date formats, status names and blank records are addressed through documented preparation controls.
- AC-11.2: Transformations do not invent missing operational facts.
- AC-11.3: Core KPIs can be reconciled to prepared data.
- AC-11.4: Portfolio outputs do not expose confidential client identity or sensitive employee/operational information.
- AC-11.5: Reconstructed logic is labeled as reconstructed where the exact historical implementation is not retained.

BRD Trace: BR-01 through BR-18; NFR-01 through NFR-06

3. Cross-Story Business Rules

Rule	Definition / Control
Completion Rate	Completed Maintenance Tasks ÷ Total Assigned Maintenance Tasks × 100
Overdue	Due Date < Current Date AND Status ≠ Completed
92% Completion	Observed dashboard finding; not a project-generated improvement
Open vs Pending	Must follow documented status mapping; do not assume equivalence
Backlog	Portfolio rebuild must define it explicitly; exact historical rule not retained
Average Completion Time	Requires valid dates and a documented date-difference convention
Workload Imbalance	Requires a documented comparison method; no historical threshold is invented
Technician Performance	Use operational measures; avoid unsupported single-score ranking

4. Acceptance Testability Matrix

Story	What Must Be Testable
US-01	Headline KPIs and completion-rate reconciliation
US-02	Status mapping and completed/open/pending totals
US-03	Due-date/status overdue rule
US-04	Valid-date completion-time calculation
US-05	Priority mapping and high-priority pending identification
US-06	Team standardization and team-level aggregation
US-07	Technician attribution and operational measures
US-08	Documented backlog/workload logic
US-09	Category and asset/equipment grouping
US-10	Filters, KPI consistency and dashboard usability
US-11	Data-quality controls, reconciliation and confidentiality

5. Definition of Done — Portfolio Reconstruction

- All confirmed business requirements are represented by one or more user stories.
- Completion and overdue rules remain identical across requirements, KPI, DAX and testing artifacts.
- Prepared data-quality controls address the confirmed source issues.
- Dashboard views support the confirmed operational analysis areas.
- Core measures can be reconciled to the prepared portfolio dataset.
- 92% is presented only as the observed overall completion rate.
- Approximately 3–4 hours/week is presented only as reporting effort eliminated.
- Unknown historical values, rankings, thresholds, code and approvals are not fabricated.
- Confidential client and sensitive operational information are excluded from public artifacts.

6. Evidence & Approval Boundary

Confirmed delivery: Power BI dashboard, live walkthrough, presentation, documentation and training.

Not claimed: These are not verbatim historical Jira stories, signed acceptance criteria, formal UAT approvals or original client ticket IDs.